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21 JANUARY 1963
SUPERSEDING
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MILITARY SPECIFICATION

PLASTIC SHEET, POLYVINYL CHLORIDE, PLASTICIZED, ELASTOMERIC

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers a thermoplastic elastomeric compound of the plasticized polyvinyl chloride resin type furnished in the form of sheets. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL–P–116 —Preservation, Methods of
MIL–P–7936 —Parts and Equipment, Aeronautical, Preparation for Delivery

STANDARDS

FEDERAL

FED–STD–406 —Plastics: Methods of Testing

MILITARY

MIL–STD–129 —Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Material. The material shall be a thermoplastic elastomeric compound of the plasticized polyvinyl chloride type, and shall conform to the applicable requirements as specified herein.

3.1.1 *Appearance.* The material shall be homogeneous and uniform in composition. The surfaces shall be smooth and free from any imperfections affecting serviceability.

3.2 Form and sizes. The polyvinyl chloride material shall be furnished in sheets. The

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dimensions and dimensional tolerances shall be as specified in the contract or order and applicable drawings.

3.3 **Physical properties.** The sheet material shall conform to the physical requirements specified in Table I, when tested as specified in 4.7.1 and 4.7.2.

3.4 **Oil immersion resistance.** The weight of the material, when tested, as specified in 4.7.3, shall not change more than ± 1.00 percent.

3.5 **Accelerated aging.** The material shall meet the requirements of Table I when aged as specified in 4.7.4.

TABLE I

Physical Properties

Property	*Requirement (1)*	*Test Paragraph*
Tensile strength	2000 psi (min.)	4.7.1
Ultimate elongation	250% (min.)	4.7.1
Hardness (Shore A)	70–85	4.7.2

(1) The material shall meet each requirement in the "as received condition" and "after oil immersion plus accelerated aging."

3.6 **Adhesion to and corrosion of metals.** When tested as specified in 4.7.5, the material shall show no tendency to adhere to either of the metallic surfaces and neither of the metallic surfaces shall show evidence of corrosion or pitting where they were in contact with the specimens. A slight discoloration of the metallic contact surface will be acceptable.

3.7 **Low temperature flexibility.** Specimens shall not break or crack when subjected to the low temperature flexibility test specified in 4.7.6.

3.8 **Workmanship.** The plastic sheet material shall be manufactured by such processes as to meet all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.**

4.2.1 *Acceptance inspection.* All inspection shall be to determine compliance with the requirements of this specification to serve as a basis for acceptance.

4.3 **Lot.** For purposes of inspection, a lot shall consist of all the sheet material manufactured at the same time, in the same production run, and offered for delivery at the same time.

4.4 **Sampling**

4.4.1 *Sampling for tests.* Three sheets of material (each at least 1½ feet square) shall be selected at random from each lot, and specimens shall be prepared from each panel.

4.5 **Examination.**

4.5.1 *Visual and dimensional examination.* Each of the samples of sheet material selected in accordance with 4.4 shall be examined to verify conformance with 3.1, 3.2 and 3.8, and all other requirements which do not involve destructive tests.

4.6 **Test conditions.** Unless otherwise specified, tests shall be conducted at a temperature of 75° ± 5° *F* (25° ± 3° C) with a relative humidity of 50 ± 5 percent. Samples shall be conditioned for at least 4 hours before testing.

4.7 **Test methods.**

4.7.1 *Tensile strength and ultimate elongation.* The tensile strength and ultimate elongation of the sheet material shall be determined as specified in Federal Test Method Standard No. 406.

4.7.2 *Hardness.* Grasp the Shore A durometer between the thumb and the third and fourth fingers, resting the index finger on top of the instrument. Place the durometer

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on the test specimen, made by laying together layers of the material to produce a thickness of not less than 0.25 inch, keeping the base of the instrument parallel to the surface of the specimen. This method of application allows the seating of the plate around the indentor flush with the test surface with a minimum amount of rocking. Apply just sufficient pressure to insure firm contact of the pressure plate with the test specimen, and make the reading immediately.

4.7.3 *Oil immersion.* Test specimens or finished parts shall be weighed and then immersed for a period of 24 hours in a paraffin base SAE 30 oil. The oil shall be maintained at a temperature of 70° F (min.) to 85° F (max.), (21° C (min.) to 29° C (max.)). After removing the specimens from the oil, they shall be allowed to drain on paper towels and then wiped dry with cheesecloth. The specimens shall be reweighed to determine the change in weight due to immersion.

4.7.4 *Accelerated aging.* After the oil immersion test, the specimens shall be subjected to dry circulated air at a temperature of 158° ± 2° F (70° ± 1° C) for 96 hours. Final determination of properties of the aged specimens shall be made not less than 20 hours after removal from the oven.

4.7.5 *Adhesion to and corrosion of metals.* Two pieces of the sample material, approximately ¾ inch square, shall be cut and placed side by side on the smooth finely-ground surface of mild steel strip 3 by 1 by ⅛ inches thick. A copper strip of the same dimensions and surface finish as the steel strip shall be placed over the sample in line with the steel strip, and with the smooth surface against the sample. The assembly of strips and samples shall be placed in a vise and a pressure of approximately 30 pounds (25 psi) applied against the outer metallic surfaces. The assembly shall then be placed in an oven and kept at 158° ± 2° F (70° ± 1° C) for 168 hours. At the end of this period, when disassembled, the samples and metallic contact surfaces shall be examined for conformance with 3.6. The material shall be considered as adhering to the metallic contact surfaces if particles or pieces of the samples remain on the metallic surfaces when the samples are removed from the metal plates.

4.7.6 *Low temperature flexibility.* Specimens in the form of strips, 4 inches long by ¼ inch wide and 1/16 inch thick shall be mounted in the form of loops over a ¾ inch square cross section board and suspended in a white gasoline bath cooled to a temperature of 0° ± 2° F (−18° ± 1° C). After 30 minutes at this temperature, the loops of the specimens shall be pinched together by hand in the bath. The hand shall be protected by using a heavy fuel-resistant glove.

4.8 Packaging, packing and marking. Preparation for delivery shall be examined for conformance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified in the contract or order, in accordance with the procedures outlined in MIL–P–7936. Level A material shall be packaged in accordance with MIL–P–116, method 1C, and if required, a separator shall be placed between each sheet.

5.2 Packing. Packing shall be level A, B or C as specified in the contract or order, in accordance with the procedures outlined in MIL–P–7936.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The elastomeric polyvinyl chloride sheet material covered by this specification is suitable for the fabrication of finished parts.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of material required (see 3.2).
(c) Quantity required.
(d) Applicable levels of packaging and packing required (see Section 5.)

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